

Foundation (Jalapeno)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 1
- Q2.** Which of the following lines is perpendicular to $y = -3x - 2$?
(A) $y = 3x + 1$
(B) $y = -\frac{1}{3}x - 2$
(C) $y = \frac{1}{3}x + 1$
(D) $y = -2x - 1$
(E) $y = 2x + 1$
- Q3.** If $y = mx + b$ and $y = -\frac{1}{m}x + c$ are perpendicular, what is $m \cdot -\frac{1}{m}$?
(A) -1
(B) 1
(C) 0
(D) 2
(E) -2
- Q4.** The equation of a line perpendicular to $y = x - 2$ is
(A) $y = -x + 1$
(B) $y = x + 2$
(C) $y = -x - 2$
(D) $y = x - 1$
(E) $y = -x + 2$
- Q5.** Which of the following is the equation of a line that is perpendicular to the line $2y + 3x = 5$?
(A) $y = -\frac{2}{3}x - 1$
(B) $y = \frac{3}{2}x + 1$
(C) $y = -\frac{3}{2}x - 2$
(D) $y = \frac{2}{3}x - 2$
(E) $y = -\frac{3}{2}x + 2$
- Q6.** If a line has a slope of m , the slope of a line perpendicular to it is $-\frac{1}{m}$.
(A) always true
(B) sometimes true
(C) never true
(D) only for lines with positive slope
(E) only for lines with negative slope
- Q7.** The equation $3y - 4x = 7$ can be written in slope-intercept form as $y =$
(A) $\frac{4}{3}x - \frac{7}{3}$
(B) $\frac{4}{3}x + \frac{7}{3}$
(C) $-\frac{4}{3}x + \frac{7}{3}$
(D) $-\frac{4}{3}x - \frac{7}{3}$
(E) $\frac{3}{4}x - \frac{7}{4}$
- Q8.** Which of the following lines is perpendicular to the line $x - 4y = 7$?
(A) $y = \frac{1}{4}x + 1$
(B) $y = -\frac{1}{4}x + 1$
(C) $y = -4x + 1$
(D) $y = 4x + 1$
(E) $y = \frac{1}{4}x - 1$
- Q9.** Write an equation of a line that passes through $(1, 2)$ and is perpendicular to $y = -2x + 3$. Show your work and explain your reasoning.
- Q10.** Find the equation of the line that is perpendicular to the line $5y - 2x = 1$ and has the same y -intercept. Explain your steps and show your calculations.

Open Ended Questions (FRQ)

- Q9.** Write an equation of a line that passes through $(1, 2)$ and is perpendicular to $y = -2x + 3$. Show your work and explain your reasoning.
- Q10.** Find the equation of the line that is perpendicular to the line $5y - 2x = 1$ and has the same y -intercept. Explain your steps and show your calculations.

Chef's Tips

- Tip 1: Ensure students understand that perpendicular lines have negative reciprocal slopes.
- Tip 2: Provide examples with various slope values to reinforce the concept.
- Tip 3: Emphasize the importance of slope in determining line relationships.